

Chapter5: DAC and ADC Part-II

Asst.Prof.Dr.Supakit Nootyaskool



Objective

- To recognize mechanism of conversion from analog signal to digital signal, with digital circuit.
- To explain a R2R circuit to convert digital data to analog signal.
- To understand concept of convert analog signal to digital signal and digital to analog signal.

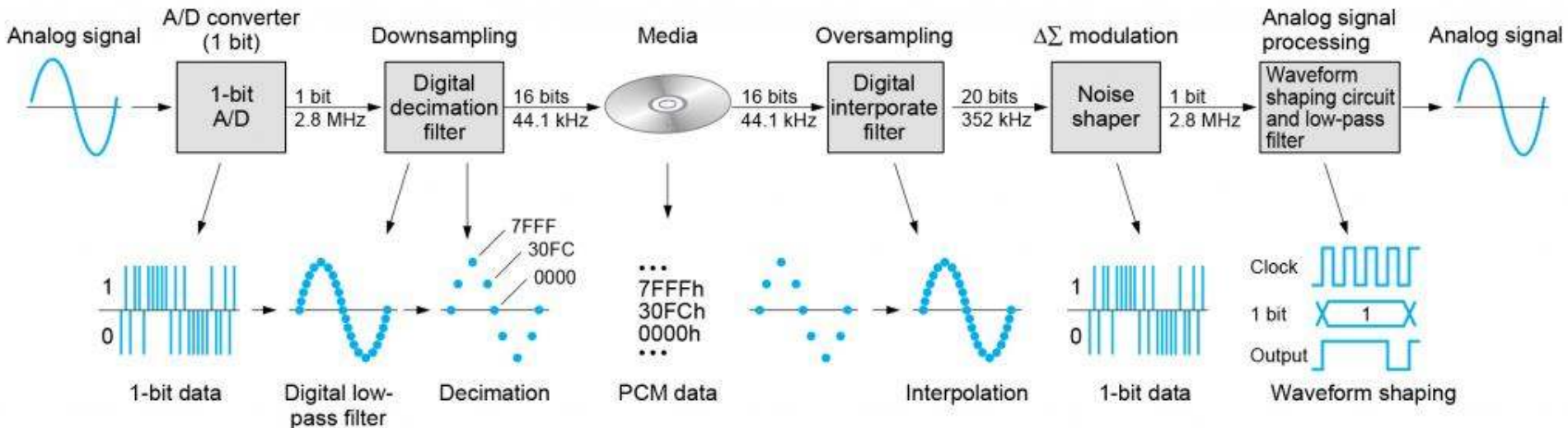
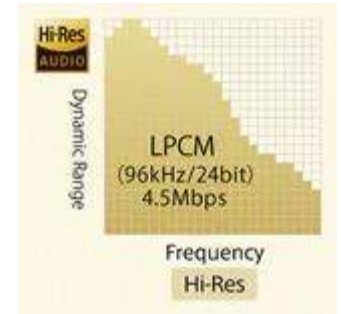
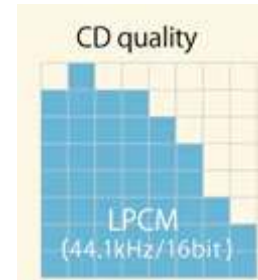
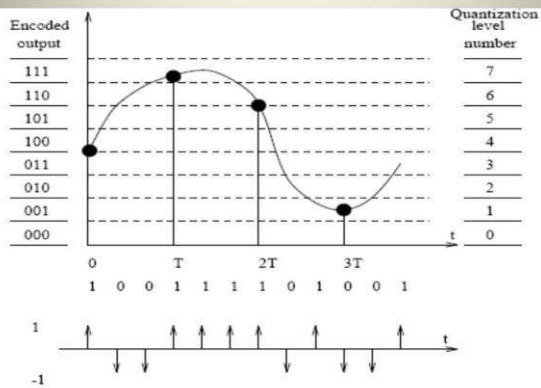
Topic

- Digital to Analog Conversion
 - R2R ladder circuit
- Analog to Digital Conversion
 - Flash ADC
 - Successive-approximations register (SAR)

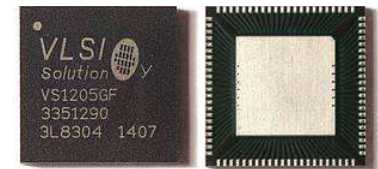
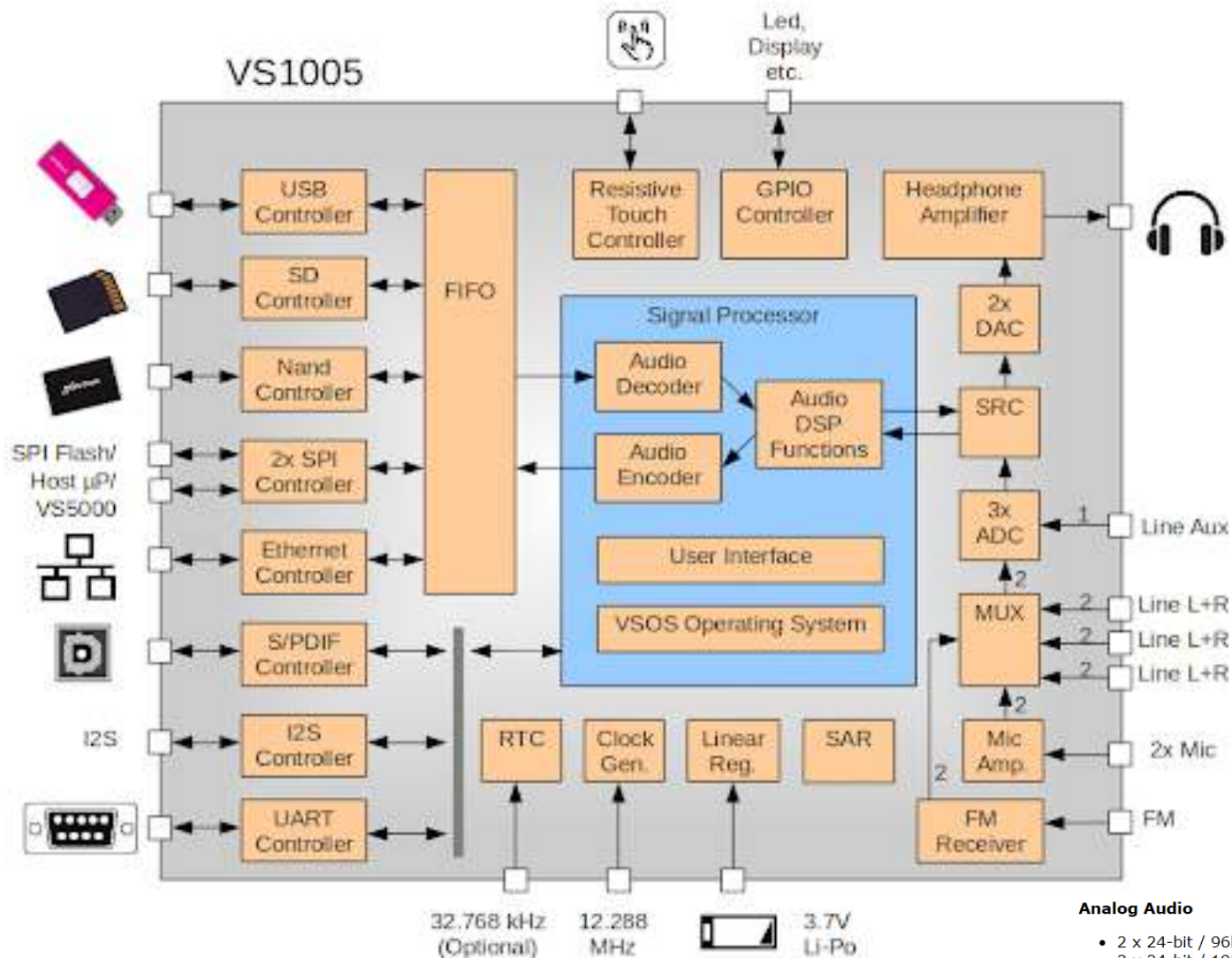
DAC AND ADC IN COMPUTER SYSTEM

Overview

Pulse Code Modulation (PCM)



MP3 Signal Processor

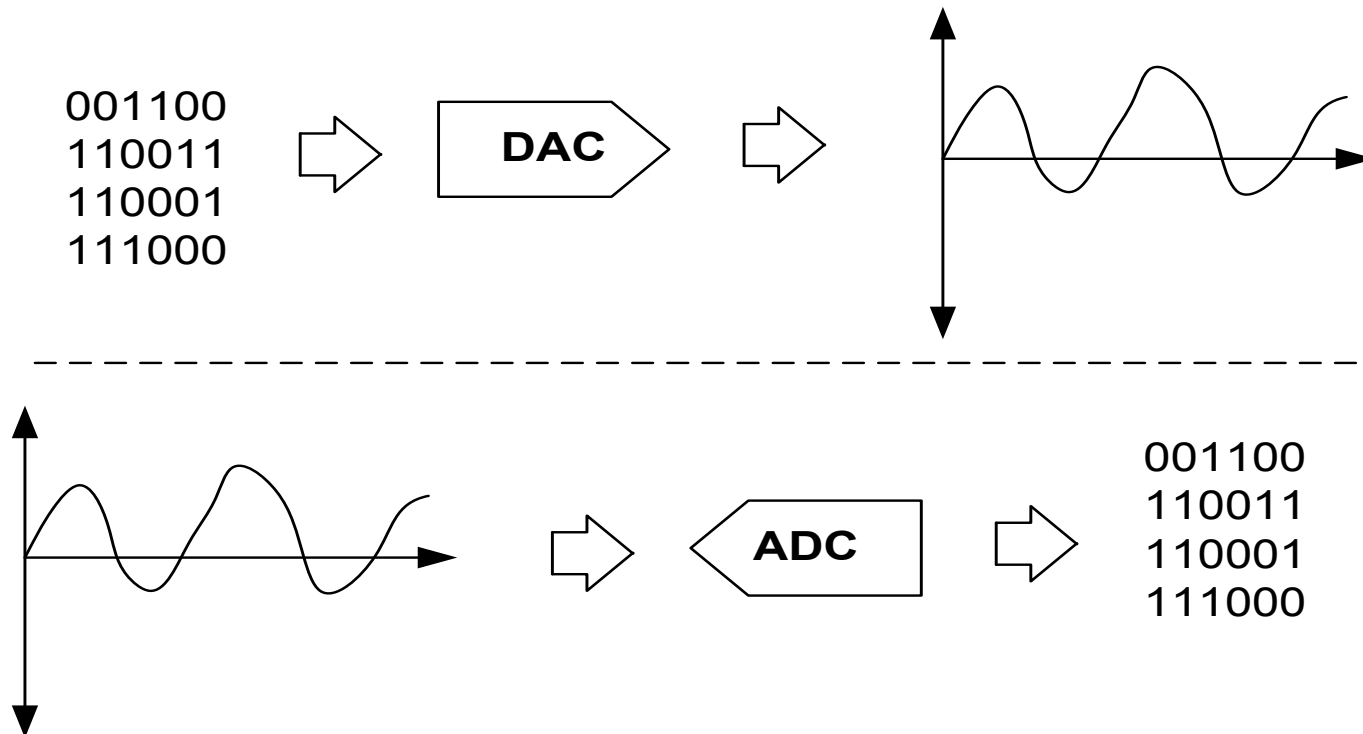


Analog Audio

- 2 x 24-bit / 96kHz DAC
- 3 x 24-bit / 192kHz ADC
- Integrated FM tuner with RDS, Japanese band included (76-108 MHz)
- Stereo headphone output
- Stereo microphone amplifier

Signal conversion concept

- DAC (Digital to Analog Conversion)
- ADC (Analog to Digital Conversion)





R2R

DIGITAL TO ANALOG CONVERSION

Example DAC design

Supposed

- 4-bits system
- Maximum output voltage at 12v

Example DAC design

$2^4 = 16$ Levels (0 to 15)

$D_0 D_1 D_2 D_3$

Supposed

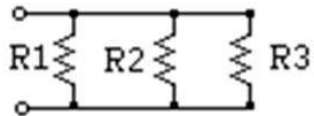
- 4-bits system
- Maximum output voltage at 12v

D3	D2	D1	D0	bitvalue	Voltage
0	0	0	0	0	0
0	0	0	1	1	0.8
0	0	1	0	2	1.6
0	0	1	1	3	2.4
0	1	0	0	4	3.2
0	1	0	1	5	4
0	1	1	0	6	4.8
0	1	1	1	7	5.6
1	0	0	0	8	6.4
1	0	0	1	9	7.2
1	0	1	0	10	8
1	0	1	1	11	8.8
1	1	0	0	12	9.6
1	1	0	1	13	10.4
1	1	1	0	14	11.2
1	1	1	1	15	12

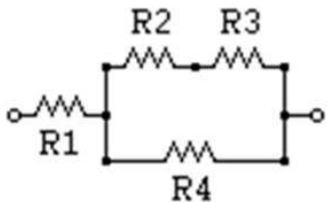
DAC: Resistance and voltage divider



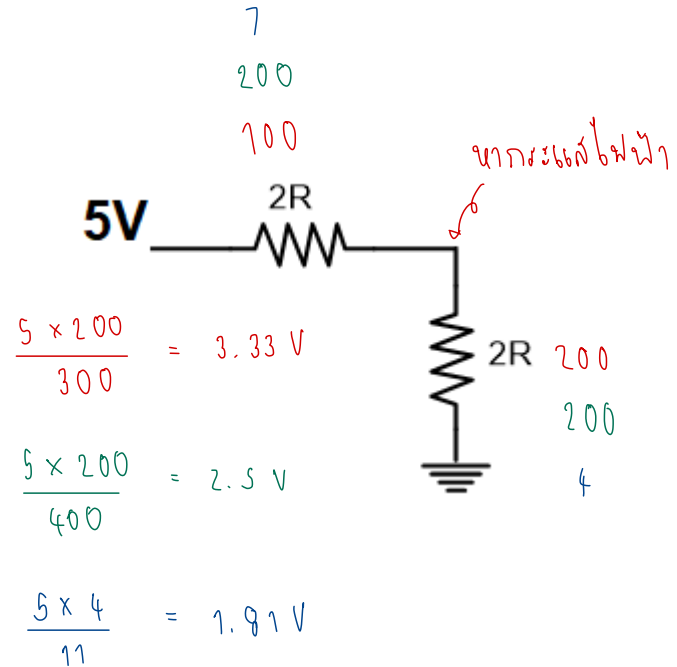
$$R_S = R1 + R2 + R3$$



$$R_P = \frac{1}{\frac{1}{R1} + \frac{1}{R2} + \frac{1}{R3}}$$



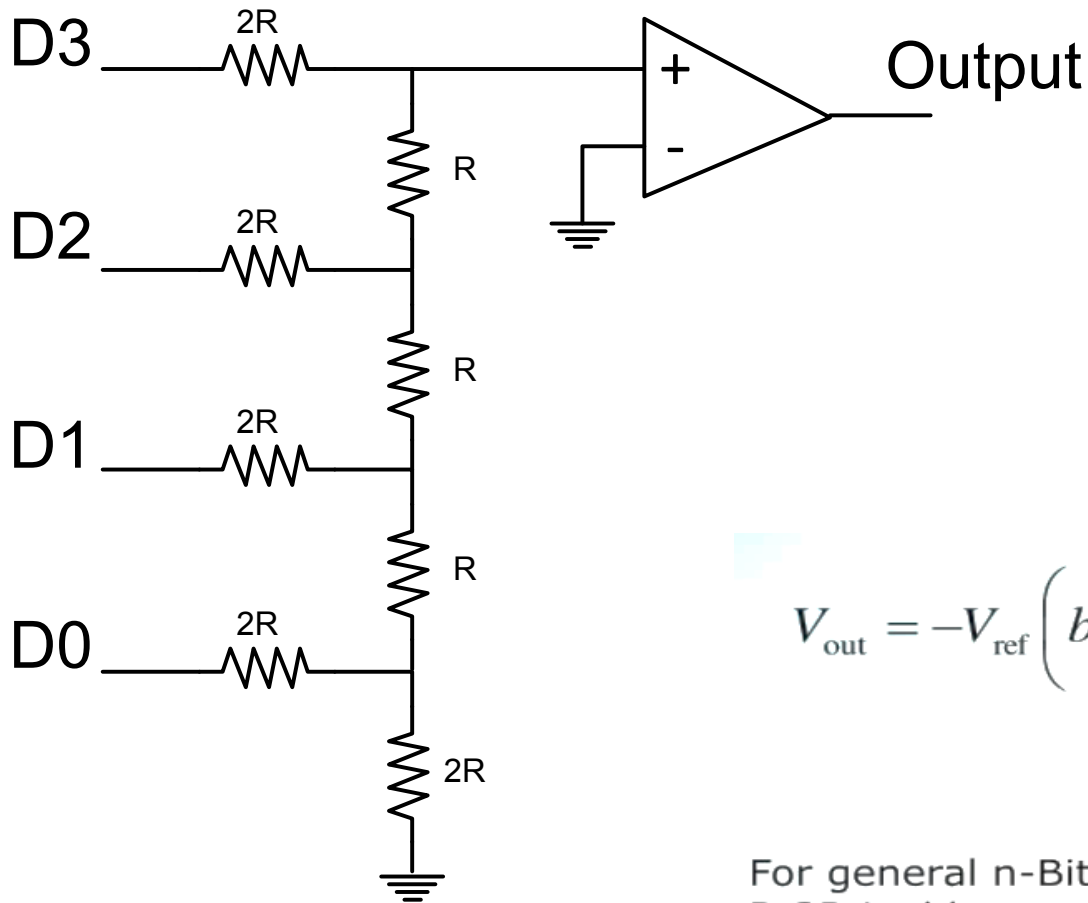
$$R = R1 + \frac{1}{\frac{1}{R2+R3} + \frac{1}{R4}}$$



$$V = I R$$

$$I = \frac{V}{R}$$

DAC circuit (R2R)



$$V_{\text{out}} = -V_{\text{ref}} \left(b_3 \frac{1}{2} + b_2 \frac{1}{4} + b_1 \frac{1}{8} + b_0 \frac{1}{16} \right)$$

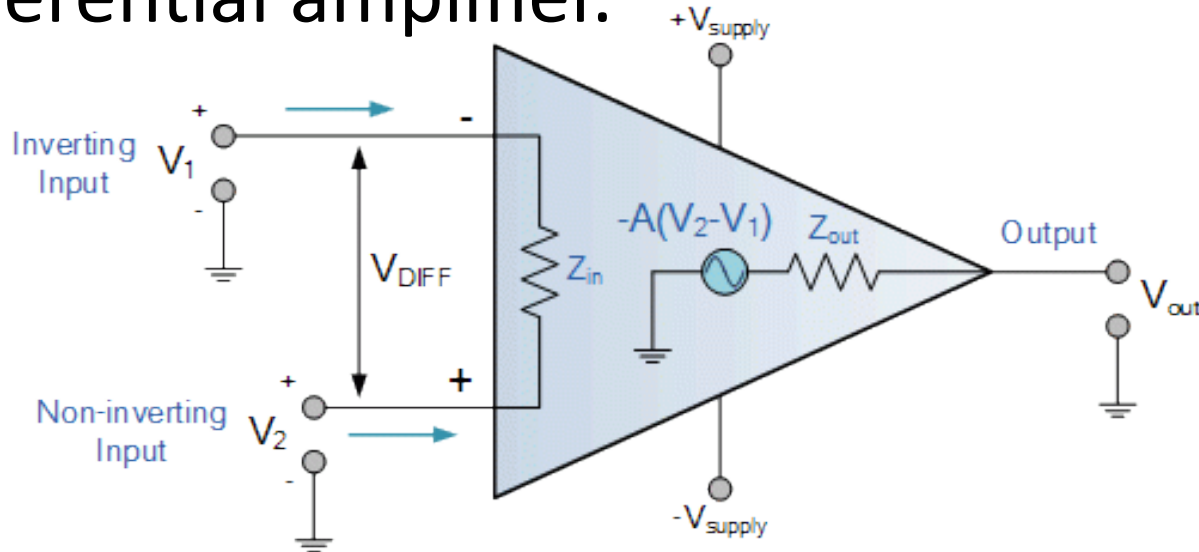
For a 4-Bit R-2R Ladder

For general n-Bit
R-2R Ladder
Binary Weighted
Resistor DAC

$$V_{\text{out}} = -V_{\text{ref}} \sum_{i=1}^n b_{n-i} \frac{1}{2^i}$$

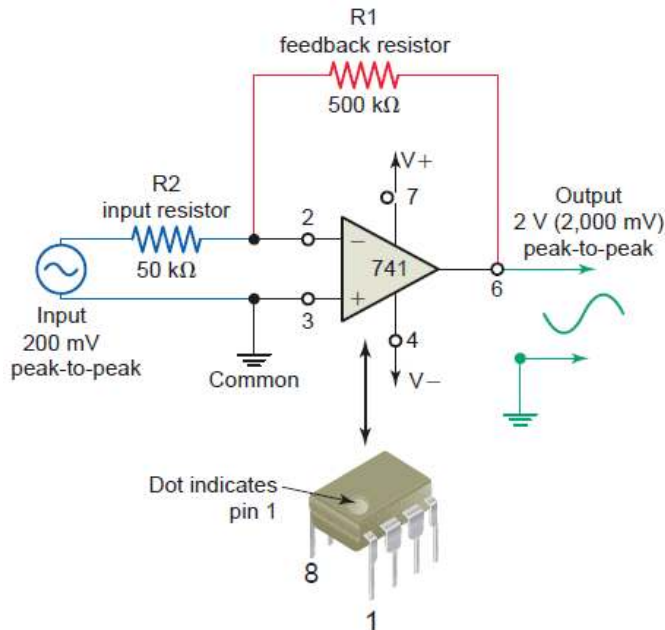
Opamp

- Operational amplifiers is an IC amplifying a small signal.
- Opamp has a very-high input impedance and having two input polarities (+) and (-).
- With input characteristic, Opamp uses for differential amplifier.



Opamp in the market

- Famous IC code
 - LM324
 - LM741
 - AD822



Single-Supply, Rail-to-Rail Low Power FET-Input Op Amp AD822-EP

FEATURES

True single-supply operation
Input voltage range extends below ground
Output swings rail-to-rail
Single-supply capability from 5 V to 30 V
Dual-supply capability from ± 2.5 V to ± 15 V
High load drive
Capacitive load drive of 350 pF, $G = +1$
Minimum output current of 15 mA
Excellent ac performance for low power
800 μ A maximum quiescent current per amplifier
Unity-gain bandwidth: 1.8 MHz
Slew rate of 3 V/ μ s
Good dc performance
800 μ V maximum input offset voltage
2 μ V/ $^{\circ}$ C typical offset voltage drift
25 pA maximum input bias current
Low noise
13 nV/ $\sqrt{\text{Hz}}$ @ 10 kHz
No phase inversion

ENHANCED PRODUCT FEATURES

Supports defense and aerospace applications (AQEC standard)
Military temperature range (-55°C to $+125^{\circ}\text{C}$)
Controlled manufacturing baseline
One assembly/test site
One fabrication site
Enhanced product change notification
Qualification data available on request

APPLICATIONS

Photodiode preamps
Active filters
12-bit to 14-bit data acquisition systems
Low power references and regulators

CONNECTION DIAGRAM

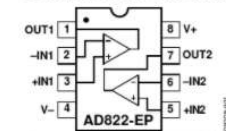


Figure 1. 8-Lead SOIC_N (R Suffix)

GENERAL DESCRIPTION

The AD822-EP is a dual precision, low power FET input op amp that can operate from a single supply of 5 V to 30 V or dual supplies of ± 2.5 V to ± 15 V. It has true single-supply capability with an input voltage range extending below the negative rail, allowing the AD822 to accommodate input signals below ground in the single-supply mode. Output voltage swing extends to within 10 mV of each rail, providing the maximum output dynamic range.

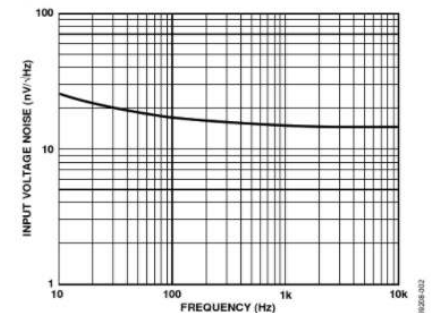


Figure 2. Input Voltage Noise vs. Frequency

Offset voltage of 800 μ V maximum, offset voltage drift of 2 μ V/ $^{\circ}$ C, input bias currents below 25 pA, and low input voltage noise provide dc precision with source impedances up to a gigaohm. The 1.8 MHz unity-gain bandwidth, -93 dB THD at 10 kHz, and 3 V/ μ s slew rate are provided with a low supply current of 800 μ A per amplifier.



TYPE OF ANALOG TO DIGITAL CIRCUIT

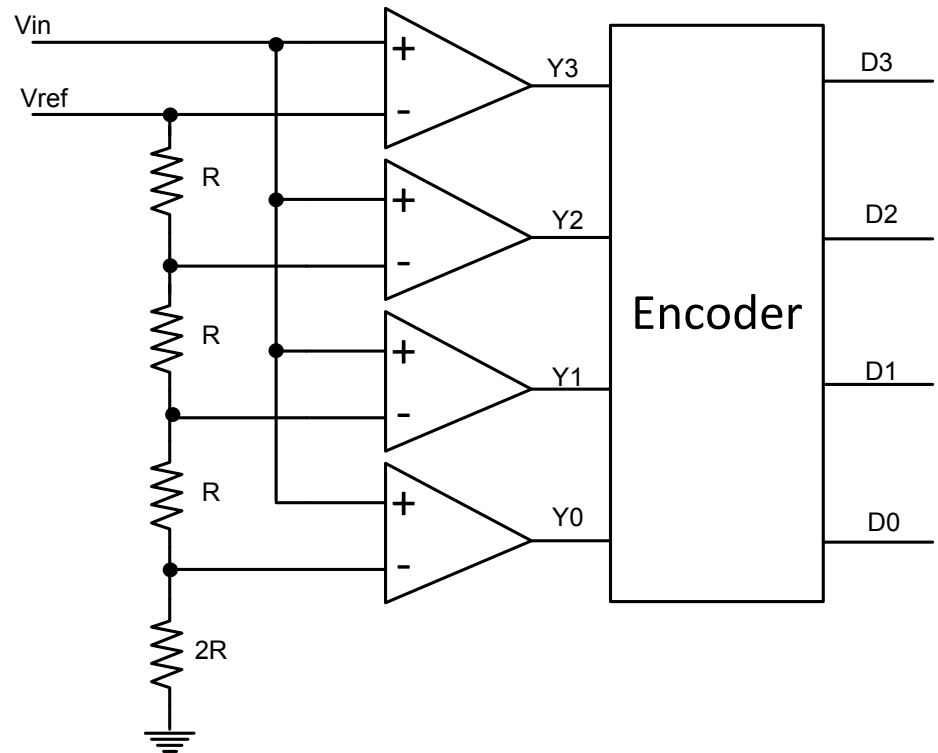
ADC circuits

- Characteristic of ADCs
 - Resolution
 - Conversion speed
- Type of ADC circuit in the present
 - Flash ADC แปลงได้เร็ว แต่แพง!
 - Pipelined ADC
 - Successive Approximation Register
 - Dual-slope convertor
 - Sigma-Delta converter

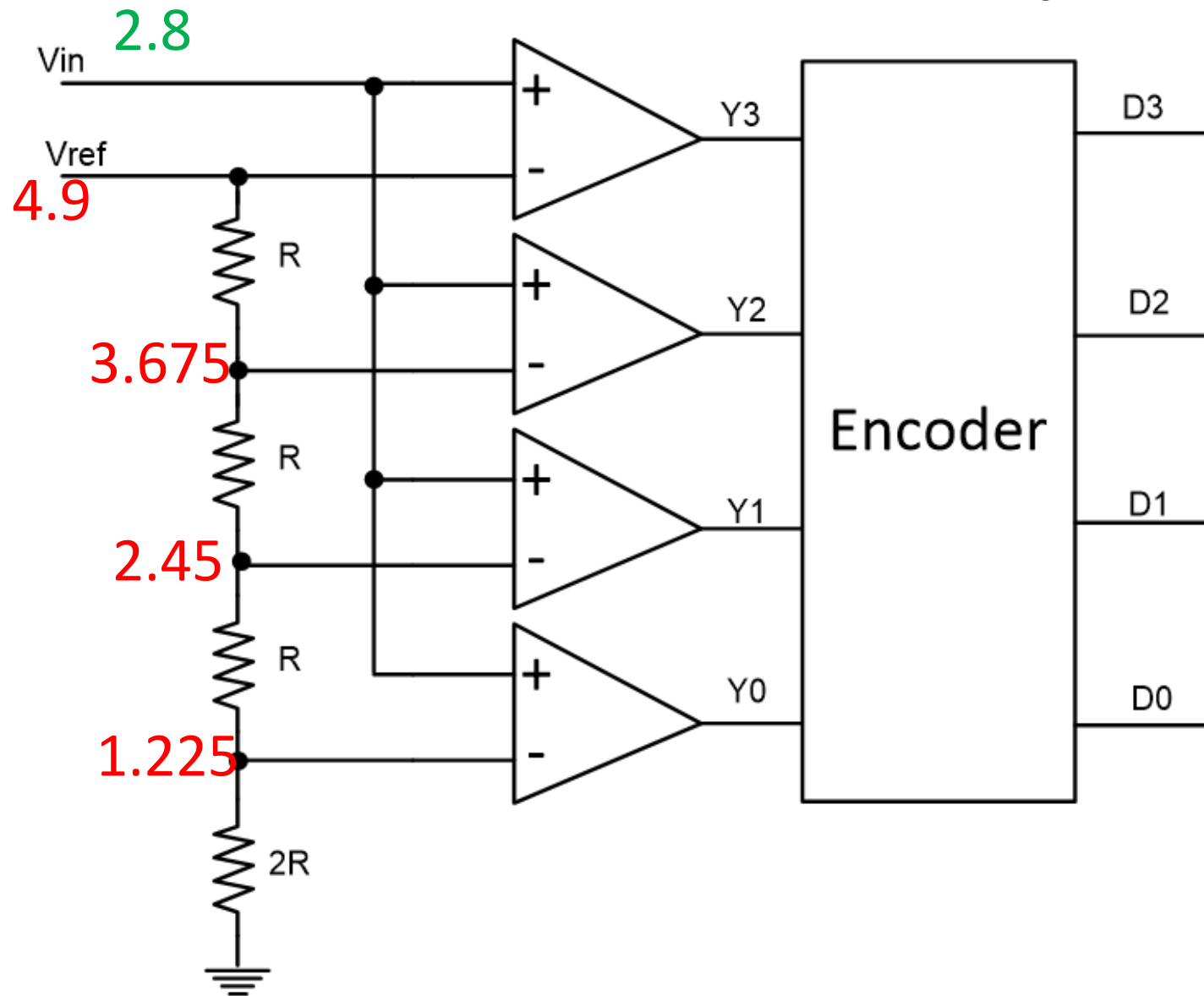
FLASH ADC

Flash ADC

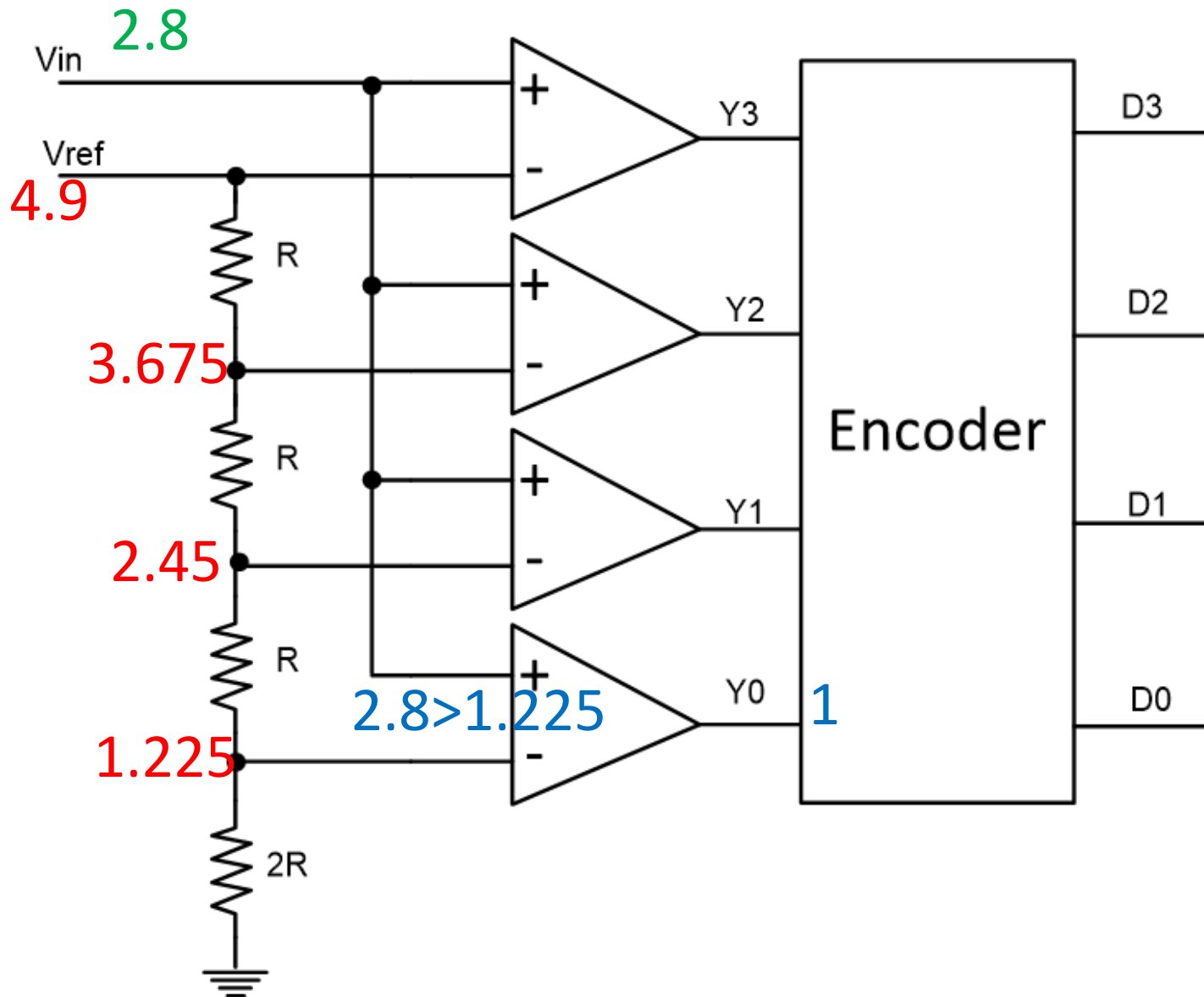
- Serie resistor for voltage dividing to create voltage references
- Compare the voltage reference with an input voltage with Opamps.
- Use encoder circuit convert to BCD data.
- Strength
 - Very fast conversion
- Weakness
 - Number of comparator require 2^{N-1}



Flash ADC: Example



Flash ADC: Example



Flash ADC: Encoder design

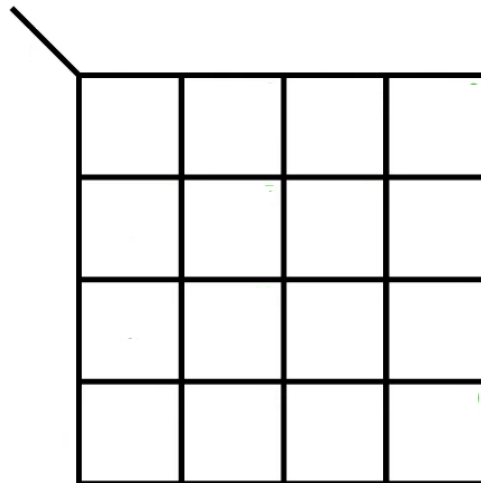
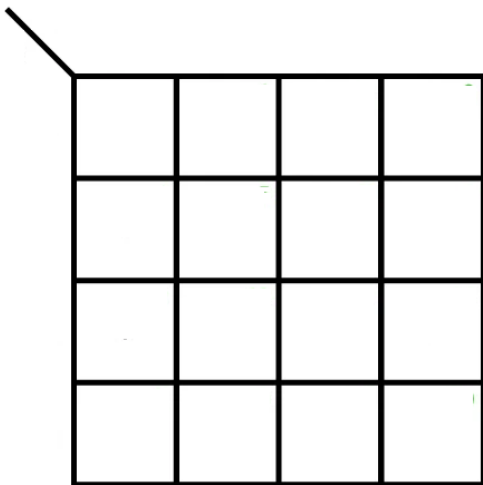
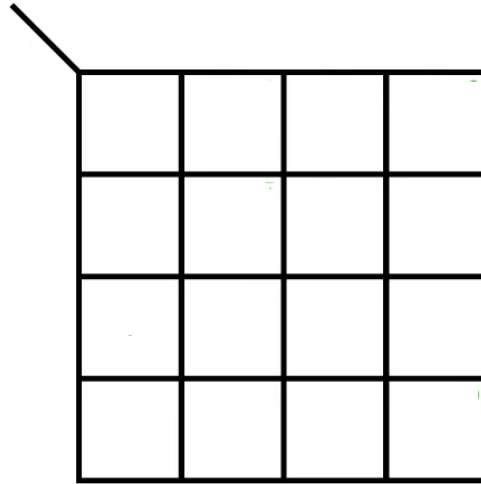
No.	Y3	Y2	Y1	Y0	D2	D1	D0
0	0	0	0	0	0	0	0
1	0	0	0	1	0	0	1
2	0	0	1	1	0	1	0
3	0	1	1	1	0	1	1
4	1	1	1	1	1	0	0

Encoder design (added don't care bits)

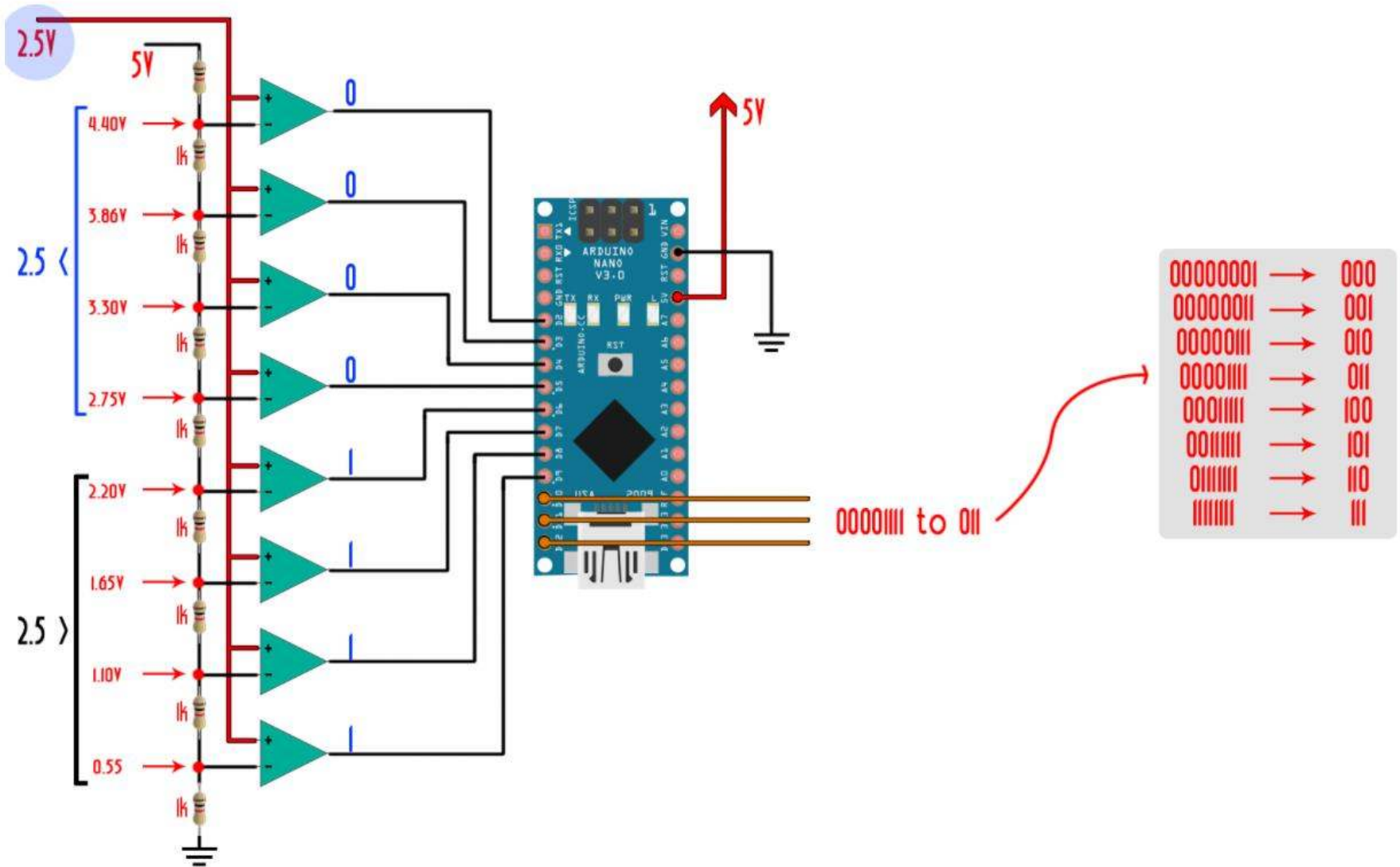
No.	Y3	Y2	Y1	Y0	D2	D1	D0
0	0	0	0	0	0	0	0
1	0	0	0	1	0	0	1
2	0	0	1	1	0	1	0
3	0	1	1	1	0	1	1
4	1	1	1	1	1	0	0
5	1	X	X	X	1	0	0
6	1	X	X	X	1	0	0
7	1	X	X	X	1	0	0

Encoder design (SOP, K-MAP)

No.	Y3	Y2	Y1	Y0	D2	D1	D0
0	0	0	0	0	0	0	0
1	0	0	0	1	0	0	1
2	0	0	1	1	0	1	0
3	0	1	1	1	0	1	1
4	1	1	1	1	1	0	0
5	1	X	X	X	1	0	0
6	1	X	X	X	1	0	0
7	1	X	X	X	1	0	0



Flash ADC applied Arduino decoder



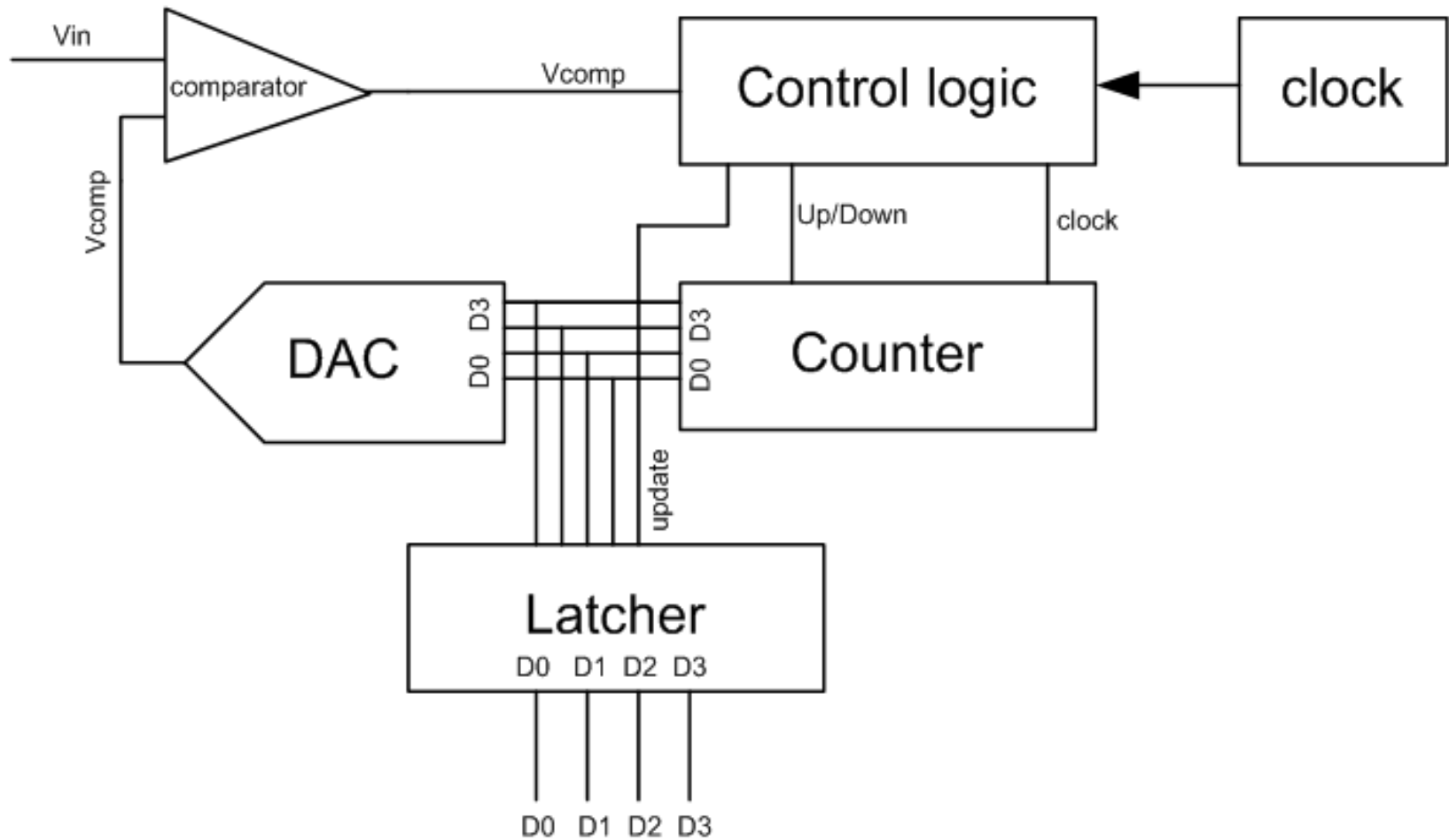
ADC: SUCCESSIVE APPROXIMATION REGISTER (SAR)

Successive Approximation Register

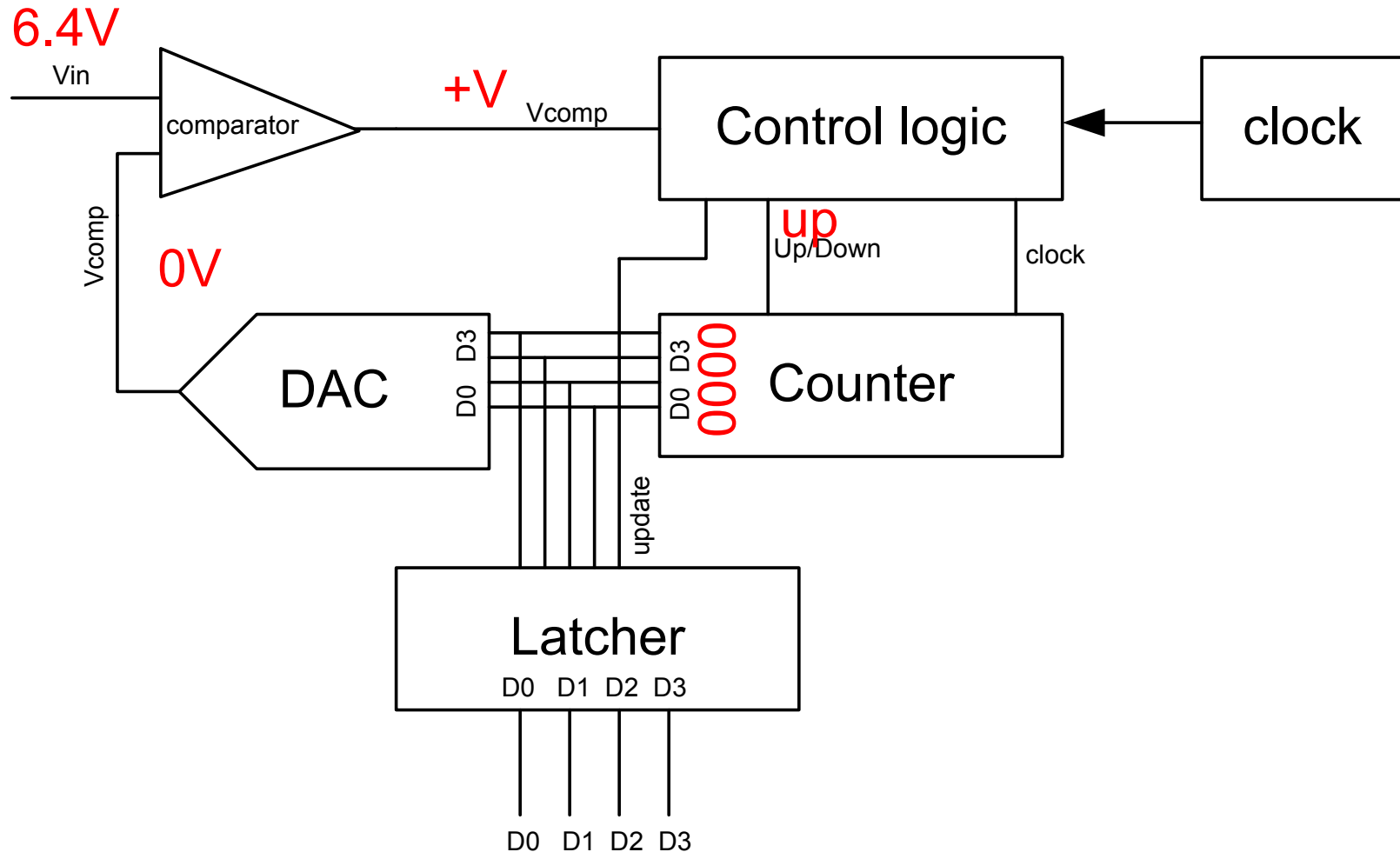
Key features

- Low energy.
- Interesting for digital circuit studying cause of having a counter, DAC, latch, and a comparator build to the system.

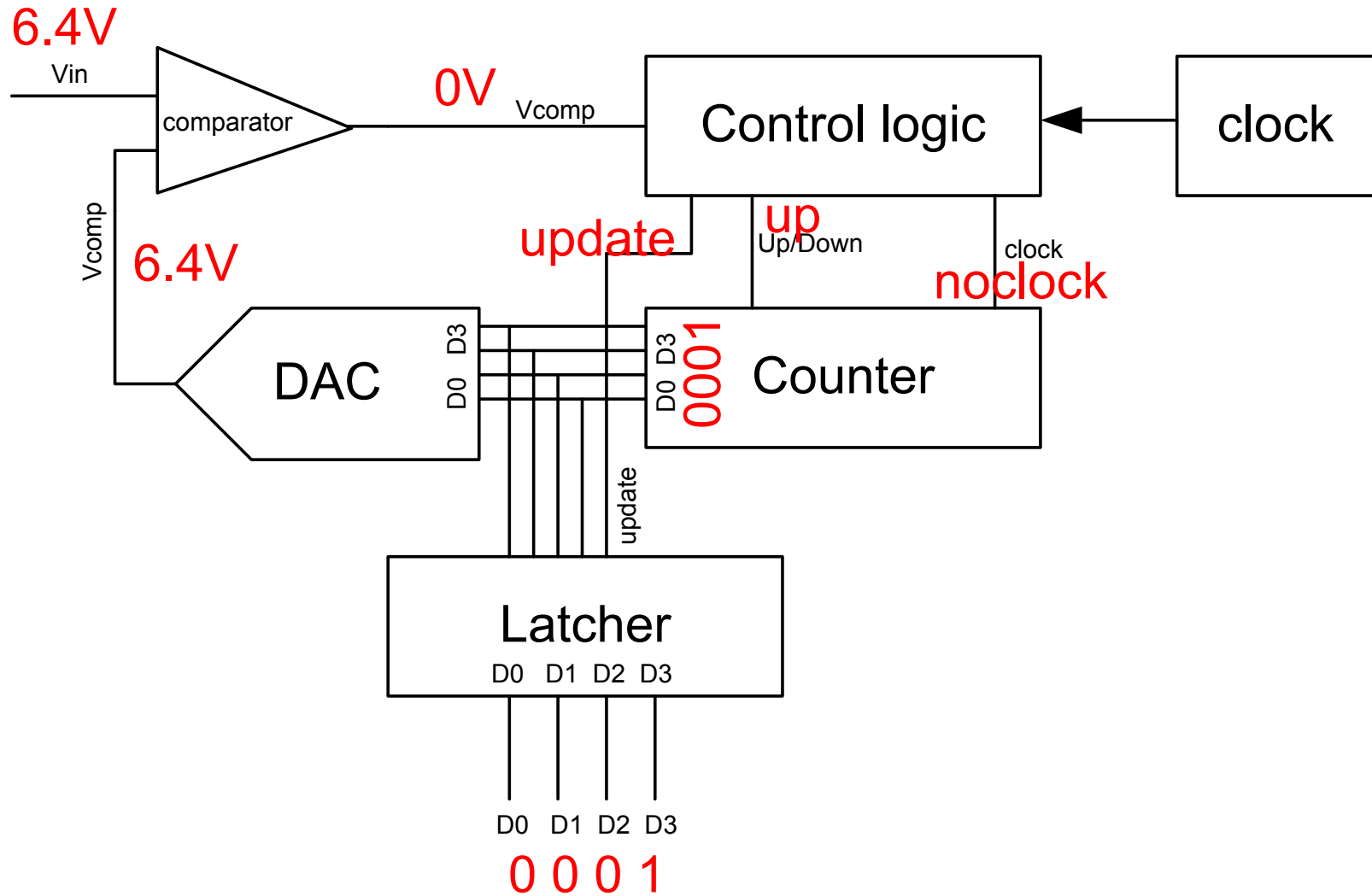
SAR-ADC



SAR-ADC: Example running



SAR-ADC: Example getting data



References

Content	Links
Tutorial digital circuit	https://www.tutorialspoint.com/digital_circuits/digital_circuits_conversion_of_flip_flops.htm
K-map online	http://www.32x8.com/index.html
Encoder circuit design	https://www.geeksforgeeks.org/encoder-in-digital-logic/
3bits flash ADC applied Arduino encoder	https://www.electrooobs.com/eng_circuitos_tut15_2.php
Flash ADC	https://www.allaboutcircuits.com/textbook/digital/chpt-13/flash-adc/
Example ADC with simulation	http://www.onmyphd.com/?p=analog.digital.converter#h3_resolution